

Computer Networks (CSC263)

Chapter-wise Study & Priority Guide

BSc. CSIT – 4th Semester – Tribhuvan University

Based on TU Exam Papers: 2076, 2078, 2079, 2080, 2080(New), 2081, 2082, Model (2077 paper not available at time of analysis)

■ ×1 Low – appeared once	■ ×2 Medium – appeared twice
■ ×3–4 High priority	■ ×5+ MUST PREPARE

★ = extremely high priority — appeared in 5+ of 8 papers, prepare thoroughly with worked/numerical examples.

Unit 1 – Introduction to Computer Network (6 Hrs)

OSI / TCP-IP Model & Layered Architecture ■ ×3

Years: 2076, 2079, 2082

Q: Explain each layer of the OSI or TCP/IP model in detail; compare the two models, or discuss the advantages of layered architecture.

Network Topologies ■ ×3

Years: 2076, 2079, 2080(New)

Q: Explain different types of network topologies (star, ring, tree, bus) along with their merits and demerits.

Network Types / Client-Server vs P2P ■ ×3

Years: 2076, 2078, 2080

Q: Explain different network types (LAN, PAN, MAN); differentiate client/server network from peer-to-peer network.

Connection-Oriented Network Services ■ ×2

Years: 2076, 2081

Q: Explain connection-oriented network services.

Protocol & Standards (definition) ■ ×1

Years: 2079

Q: Define protocol; explain why we need standards.

■ **What to read:** Foundational unit — mostly definitional, spread thin across several sub-topics rather than one dominant question. Know the OSI/TCP-IP layer functions cold since they get referenced throughout the rest of the paper (e.g. "where does this protocol reside in the OSI layer" shows up in Unit 3 questions too).

Unit 2 – Physical Layer and Network Media (4 Hrs)

Switching (Circuit / Packet / Message) ■ ×5 ★

Years: 2076, 2079, 2080, 2080(New), Model

Q: Explain circuit switching or packet switching with its advantages and disadvantages; compare circuit-switched and packet-switched networks.

Transmission Media (guided, unguided, wireless) ■ ×4 ★

Years: 2076, 2078, 2081, 2082

Q: Differentiate guided from unguided transmission media; explain any two wireless transmission media in detail.

Network Devices & OSI Layer Mapping / LAN Design ■ ×2

Years: 2076, Model

Q: Identify which OSI layer a given device (Hub, Switch, Router) operates in; design a LAN for a given office scenario, justifying the connecting devices chosen.

ISDN & Ethernet Cable Standards ■ ×1

Years: 2078

Q: Write a short note on ISDN interface and standards.

■ **What to read:** Switching is the anchor topic — always paired with advantages/disadvantages or a compare/contrast question, so have both circuit and packet switching definitions plus a clean comparison table ready. Transmission media is usually a quick differentiate-or-list question.

Unit 3 – Data Link Layer (8 Hrs)

MAC / LLC / Wireless LAN / Token Ring-Bus / CSMA ■ ×6 ★

Years: 2076, 2078, 2080, 2080(New), 2081, Model

Q: Illustrate the functionality of Media Access Control; explain CSMA/CD, Token Bus/Token Ring, or the architecture of IEEE 802.11 wireless LAN.

Error Detection: CRC / Hamming Code / Checksum ■ ×5 ★

Years: 2076, 2079, 2081, 2082, Model

Q: Explain CRC (Cyclic Redundancy Check) with a numerical example, or find the Hamming code for given data; differentiate error detection from error correction.

Flow Control & ARQ (Stop-and-Wait / Go-Back-N / Selective Repeat) ■ ×4 ★

Years: 2076, 2078, 2080(New), 2082

Q: Define flow control; explain Stop-and-Wait ARQ or Go-Back-N ARQ with an example, and how it differs from the other.

Virtual Circuit Networks / Frame Relay / ATM ■ ×4 ★

Years: 2078, 2079, 2082, Model

Q: Explain what a virtual circuit network is; describe ATM or Frame Relay as a virtual-circuit wide area network, or differentiate the two.

PPP / HDLC ■ ×3

Years: 2080, 2080(New), 2081

Q: Explain the PPP protocol and its link setup process; state the functionality of Logical Link Control and briefly explain HDLC.

■ **What to read:** Heaviest unit by hours (8) and it earns it — four different topics all clear the high-priority threshold. CRC/Hamming numericals are guaranteed marks if you practice the calculation steps by hand. MAC/wireless LAN and flow-control ARQ protocols are the other two pillars — all four deserve equal attention.

Unit 4 – Network Layer (10 Hrs)

IPv4/IPv6 Addressing & Subnetting Numerical ■ ×7 ★

Years: 2076, 2078, 2079, 2080, 2080(New), 2081, 2082

Q: Given a network address (class A/B/C), divide it into a specified number of subnets and find the new subnet mask, subnet ID, broadcast address and address range for each.

Routing (Distance Vector / Link State / Routing Table) ■ ×7 ★

Years: 2076, 2078, 2079, 2080, 2080(New), 2082, Model

Q: Explain Distance Vector Routing or Link State Routing with a suitable example; differentiate static from dynamic routing tables.

IPv4 vs IPv6 (comparison, datagram format, transition reasons) ■ ×6 ★

Years: 2078, 2080, 2080(New), 2081, 2082, Model

Q: Differentiate IPv4 from IPv6 protocol; describe the IPv6 datagram format and give reasons for the transition from IPv4 to IPv6.

NAT / ICMP / IGMP ■ ×2

Years: 2076, 2081

Q: Explain what NAT is, how it works and its benefits; write short notes on ICMP and IGMP.

Network Traffic Analysis / Firewall / Access Control ■ ×1

Years: Model

Q: Discuss why network traffic analysis is carried out; write a short note on firewalls.

■ **What to read:** The single most important unit in the whole subject — two topics tied at ×7/8 papers each. Subnetting numericals are 100% guaranteed every year — practice class A/B/C subnetting until you can compute subnet ID, broadcast address and host range without hesitation. Pair it with Distance Vector/Link State routing examples, since both are asked as long-answer questions almost every time.

Unit 5 – Transport Layer (6 Hrs)

Traffic Shaping Algorithms (Leaky Bucket / Token Bucket) ■ ×4 ★

Years: 2080, 2080(New), 2081, 2082

Q: Explain the working mechanism of the leaky bucket or token bucket algorithm with an example.

TCP Header / 3-Way Handshake / TCP vs UDP ■ ×4 ★

Years: 2079, 2080, 2080(New), 2082

Q: Explain the TCP header with a neat diagram and its uses; describe how a TCP connection is established using 3-way handshaking, or differentiate TCP from UDP.

Socket Programming / Ports ■ ×3

Years: 2080, 2080(New), Model

Q: Differentiate port from socket; demonstrate the use of socket programming for creating a network application using UDP and TCP.

Congestion Control (Open-Loop vs Closed-Loop) ■ ×2

Years: 2078, Model

Q: Explain open-loop congestion control and compare it with closed-loop congestion control approaches.

■ **What to read:** Traffic shaping and TCP fundamentals (header/handshake/vs UDP) are your two safest bets here — both are asked almost every year in some form. Socket programming questions usually want a diagram plus a short code-flow description, not full implementation — don't over-prepare the coding side.

Unit 6 – Application Layer (7 Hrs)

DNS (purpose, resolution, query types, hierarchy) ■ ×5 ★

Years: 2076, 2080(New), 2081, 2082, Model

Q: Explain the major purpose of DNS and how it resolves a domain name to an IP address; discuss recursive, non-recursive and iterative DNS queries, or DNS hierarchy and record types.

FTP / SFTP / Email Protocols (SMTP, IMAP, POP3, SNMP) ■ ×3

Years: 2080, 2080(New), 2082

Q: Explain FTP and SFTP; differentiate IMAP from POP3; explain SNMP.

HTTP / WWW / URL ■ ×2

Years: 2079, 2082

Q: Explain what HTTP protocol is, covering HTTP request and response messages; describe the architecture of WWW and define URL.

Proxy / Web / Mail Server ■ ×2

Years: 2081, Model

Q: Differentiate web server from proxy server; explain the function of a proxy server and electronic mail.

■ **What to read:** DNS is the clear anchor topic here — know the resolution process end-to-end plus the three query types (recursive/non-recursive/iterative). The rest of this unit is a grab-bag of shorter definitional questions (HTTP, FTP, mail protocols) — keep concise one-paragraph answers ready for each rather than deep-diving any single one.

Unit 7 – Multimedia & Future Networking (4 Hrs)

SDN / NFV / NGN ■ ×6 ★

Years: 2076, 2079, 2080, 2080(New), 2081, 2082

Q: Explain what a Software Defined Network (SDN) is and its architecture; discuss why Network Functions Virtualization (NFV) is required, or describe the architecture of NGN.

Multimedia Streaming Protocols ■ ×1

Years: 2078

Q: Discuss the different approaches for multimedia streaming.

■ **What to read:** Small unit but SDN/NFV/NGN punches well above its weight — appears in 6 of 8 papers, almost always as a short-to-medium question. Know the SDN architecture (control plane vs data plane separation) and one-line definitions of NFV and NGN — that covers nearly everything asked here.

Top Priority Summary (Sorted by Frequency)

Topic	Unit	Freq	Years
IPv4/IPv6 Addressing & Subnetting Numerical	Network Layer	×7	2076–82 (7 papers)
Routing (Distance Vector/Link State/Table)	Network Layer	×7	2076–80(N),82,Model
IPv4 vs IPv6 (comparison/datagram/transition)	Network Layer	×6	2078,80,80(N),81,82,Model
SDN / NFV / NGN	Multimedia & Future Networking	×6	2076,79,80,80(N),81,82
MAC/LLC/Wireless LAN/Token Ring-Bus	Data Link Layer	×6	2076,78,80,80(N),81,Model
Switching (Circuit/Packet/Message)	Physical Layer & Network Media	×5	2076,79,80,80(N),Model
Error Detection (CRC/Hamming/Checksum)	Data Link Layer	×5	2076,79,81,82,Model
DNS (purpose/resolution/query types)	Application Layer	×5	2076,80(N),81,82,Model
Flow Control & ARQ (Stop-Wait/Go-Back-N)	Data Link Layer	×4	2076,78,80(N),82
Virtual Circuit/Frame Relay/ATM	Data Link Layer	×4	2078,79,82,Model
Transmission Media (guided/unguided/wireless)	Physical Layer & Network Media	×4	2076,78,81,82
Traffic Shaping (Leaky/Token Bucket)	Transport Layer	×4	2080,80(N),81,82
TCP Header / 3-Way Handshake / TCP vs UDP	Transport Layer	×4	2079,80,80(N),82
OSI/TCP-IP Model & Layered Architecture	Introduction	×3	2076,79,82
Network Topologies	Introduction	×3	2076,79,80(N)
Network Types / Client-Server vs P2P	Introduction	×3	2076,78,80
PPP / HDLC	Data Link Layer	×3	2080,80(N),81
Socket Programming / Ports	Transport Layer	×3	2080,80(N),Model
FTP/SFTP/Email Protocols	Application Layer	×3	2080,80(N),82

Final Exam Very High Probability Topics (Never Miss These)

MUST PREPARE (×5+, near-certain): IPv4/IPv6 subnetting numerical, Routing (DV/link state/table), IPv4 vs IPv6, SDN/NFV/NGN, MAC/wireless LAN, Switching, CRC/Hamming/checksum, DNS.

High priority (×3–4): Flow control & ARQ, Virtual circuit/Frame Relay/ATM, Transmission media, Traffic shaping, TCP header/handshake/vs UDP, OSI/TCP-IP model, Network topologies, Network types, PPP/HDLC, Socket programming, FTP/email protocols.

Numerical-heavy (practice by hand every time): Subnetting (subnet ID, broadcast address, host range), CRC calculation, Hamming code, network/broadcast address from a given IP/prefix.

Due for a repeat (asked before, quiet recently — good bet): NAT/ICMP/IGMP, Network traffic analysis/firewall, Congestion control, Proxy/web/mail server, HTTP/WWW/URL, Multimedia streaming, ISDN.